

Amrita School of Computing, Amritapuri
Department of Computer Science and Engineering(AI)
B.Tech 2022-2026 CSE(AI)
22AIE499 Project Phase 2 -Zeroth Proposal

1. Project Group No: 5

2. Project Title:

Implementing and Evaluating Neural and Large Language Models for Sanskrit
Sandhi Splitting in Ayurvedic Sanskrit Texts

3. Team Members:

Name	Reg No
Abhiram A S	AM.EN.U4AIE22002
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4. Problem Statement:

The objective of Phase 1 was to study and evaluate neural approaches for Sanskrit Sandhi splitting, a core preprocessing task in Sanskrit NLP. Phase 1 involved a linguistic study of Sandhi, a review of rule-based and neural approaches, and the implementation of character-level neural models including sequence-to-sequence and two-stage architectures. The models were evaluated using standard metrics, and detailed error analysis was performed. This study highlighted key limitations of purely neural methods, particularly under low-resource settings and in handling Sanskrit's rule-governed grammatical structure.

Based on the feedback provided by the review panel, Phase 2 focuses on the implementation and systematic evaluation of neural models and modern LLMs for Sanskrit Sandhi splitting. The objectives of Phase 2 are as follows:

- Systematic benchmarking of state-of-the-art multilingual LLMs and low-resource models for Sanskrit.
- Cross-language comparison to highlight Sanskrit's unique grammatical and linguistic characteristics.
- Detailed qualitative and quantitative error analysis to identify model strengths and failure cases.
- A dedicated module exploring Panini-inspired rule-based or hybrid approaches to address the limitations observed in purely neural and LLM-based models.

5. Phase 1 Alpha paper publication Status:

Not Published

Guide's Name: Jayasree Nair

Signature:

A handwritten signature in black ink, appearing to be 'Jayasree Nair', written in a cursive style.

Date: 22 - 01 - 2026